Restoring mutant Huntingtin conformational flexibility
Identification of small molecule modifiers using TR-FRET

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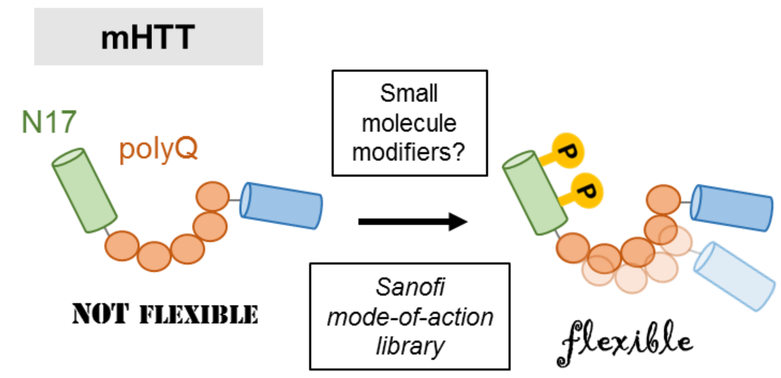
INTRODUCTION

- Novel approach to treat Huntington's disease (HD) by restoring mutant Huntingtin's (mHTT) conformational flexibility
- Rescue of mHTT conformational flexibility has been attributed to lower levels of aggregation and neurotoxicity in animal models
- Time-resolved Förster Resonance Energy Transfer (TR-FRET) can detect HTT flexibility state changes due to mHTT's altered reaction to temperature change
- Screen for small molecules that modify mHTT flexibility through posttranslational modifications (PTMs) in human patient fibroblasts currently ongoing

RATIONAL: REPAIRING mHTT INFLEXIBILITY

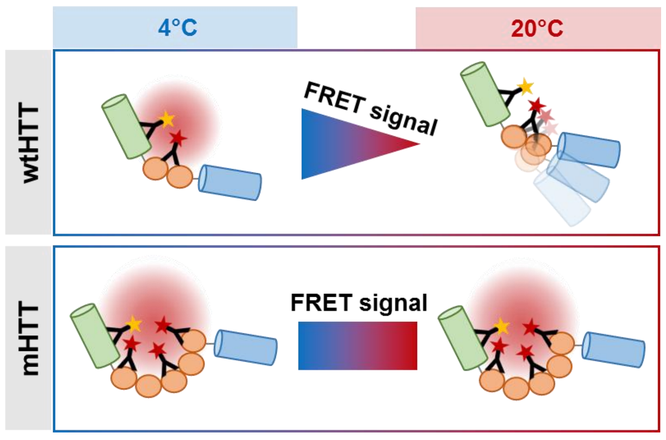


mHTT N-terminus is **structured & less flexible**, leading to **aggregation & loss of function.**

N-terminal **post-translational modifications (PTMs) can revert mHTT inflexibility.**

Novel targets for HD therapeutics?

APPROACH: TEMPERATURE-SENSITIVE mHTT TR-FRET

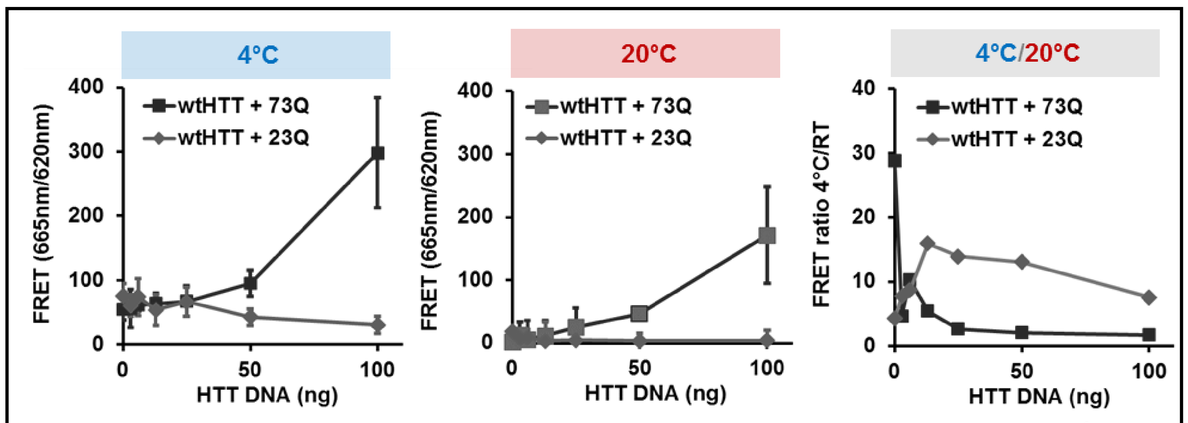


Temperature influences HTT TR-FRET signals (MW1/2B7 antibody pair) due to protein flexibility & epitope accessibility.

When switching temperatures, the **relative FRET signals of flexible wtHTT and inflexible mHTT differ.**

Fodale *et al.*, PLoS One, 2014
Cui *et al.*, Sci Rep, 2014
Daldin *et al.*, Sci Rep, 2017

OVEREXPRESSED HTT TR-FRET IN COMPLEX CELL LYSATES

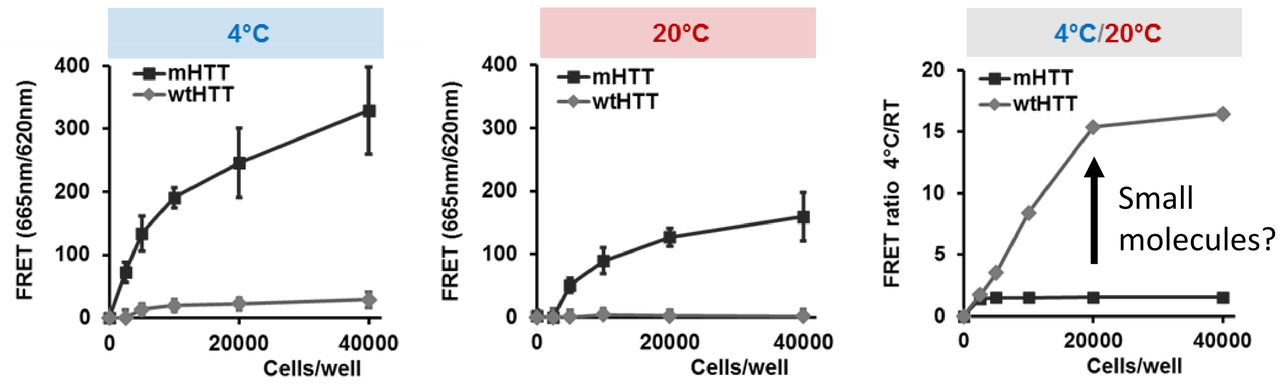


Transfected 73Q HTT yields higher TR-FRET signals than 23Q DNA in wtHTT cells. **The assay is therefore specific for mHTT** even in the presence of complex cell lysate.

At 20°C both the 73Q and 23Q HTT signals are reduced due to **increased molecular movement.**

Calculating the **4°C/20°C ratio** indicates that **23Q HTT is relatively more flexible** than 73Q HTT, mainly due to high flexibility at 20°C and accordingly strongly decreased FRET signal.

ENDOGNEOUS HTT TR-FRET IN PATIENT-DERIVED FIBROBLAST LYSATES



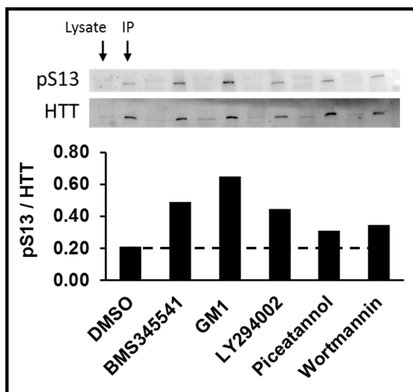
Absolute FRET signals higher for mHTT due to higher MW1 epitope number. At **4°C HTT flexibility is decreased** and FRET signals are higher overall.

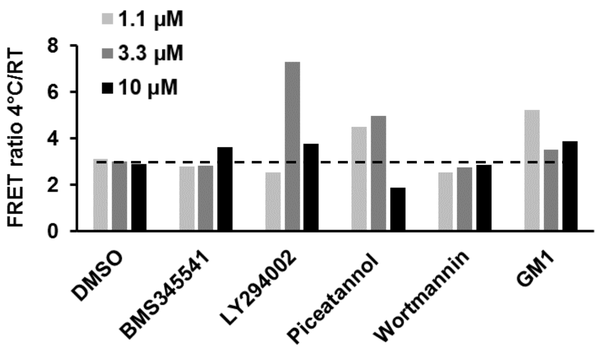
At **20°C HTT flexibility is increased** and FRET signals are lower overall.

The reaction to temperature changes differs between wtHTT and mHTT, due to relatively higher wtHTT flexibility.

*The **4°C/20°C fold change** is higher for wtHTT due to its **higher flexibility at 20°C** .*

SEVERAL PUBLISHED COMPOUNDS INCREASE pS13 LEVELS & mHTT FLEXIBILITY



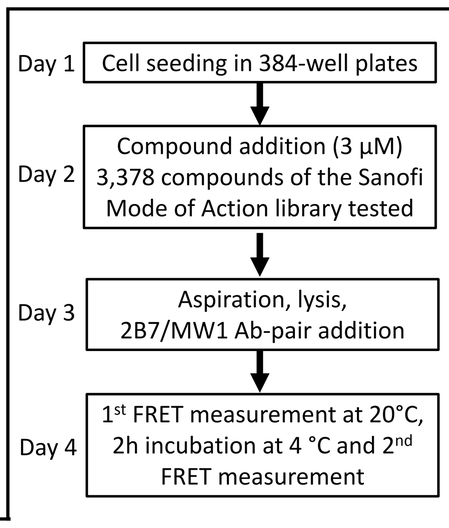


Are mHTT N17 PTMs a general mechanism to regulate conformational flexibility?

As described in the literature, several compounds mildly increase HTT pS13 and some also increase mHTT flexibility in a TR-FRET assay.

Atwal *et al.*, Nat Chem Biol, 2011
Di Pardo *et al.*, PNAS, 2012
Bowie *et al.*, PNAS, 2018

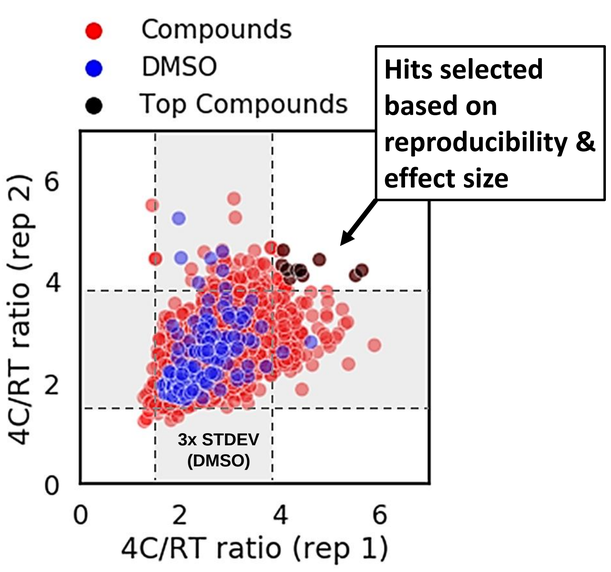
ONGOING SCREEN FOR mHTT FLEXIBILITY MODIFIERS

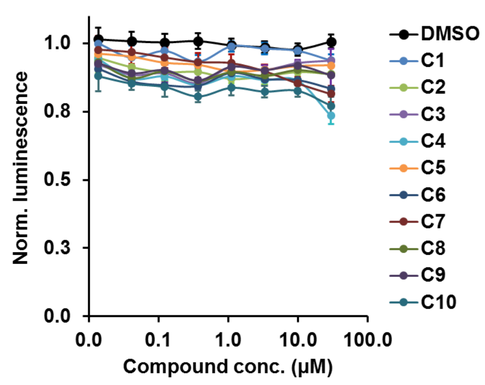


- At 20°C we **identified several HSP90 inhibitors** known to reduce mHTT levels

Orozco-Díaz *et al.*, PLoS One, 2019
Baldo *et al.*, JBC, 2012

- FRET signal reduction can be due to:
  1) Cellular toxicity
  2) mHTT reduction
  3) Increased mHTT flexibility





An ATP assay indicates no strong toxicity for the top 10 hits.

NEXT STEPS

- Continued screening of the Mode-of-Action library
- Toxicity evaluation & dose-response testing of all 4°C/20°C FRET ratio modifying compounds
- Mass-spectrometry analysis of positive hits to confirm known pT3, pS13, pS16 modifications or to identify novel PTMs

Screen to be continued

TECHNICAL SUPPORT & ADVICE

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